ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C8 · SHADOWS

Light, materials and shadows

★☆☆
Grade 5

Learning goals

- Classify materials.
- Recall how shadows form and change.

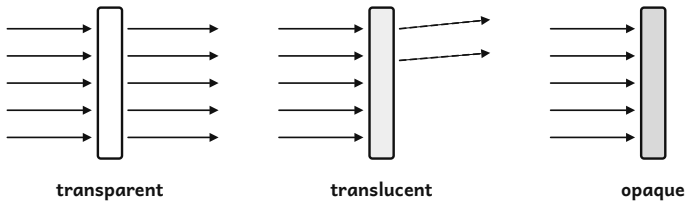
1 **Sort the materials.** [9]

SORT THESE

clear plastic bottle **tracing paper** **water** **clear glass** **steel tray** **cardboard** **wood** **thin cotton cloth** **frosted glass**

Transparent	Translucent	Opaque
clear glass	tracing paper	wood
water	frosted glass	cardboard
clear plastic bottle	thin cotton cloth	steel tray

2 **Why can you read a book through clear glass but not through tracing paper or cardboard?** [3]



Glass is transparent: light passes straight through. Tracing paper is translucent: it scatters the light so the words look blurred. Cardboard is opaque: no light gets through.

3 **Fill in the gaps.** [5]

WORD BANK

longest **lux** **opaque** **curved** **metres** **straight** **shortest**

Light travels in straight lines. A shadow forms when an opaque object blocks the light. Shadows are longest in the morning and evening and shortest at midday. Light intensity is measured in lux.

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. A shadow shows the colours of the object.		✔	A shadow only shows the outline shape.
2. Moving an object closer to the torch makes its shadow bigger.	✔		
3. Shadows point towards the Sun.		✔	Shadows point away from the Sun.
4. A light meter measures how bright light is.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C8 · SHADOWS

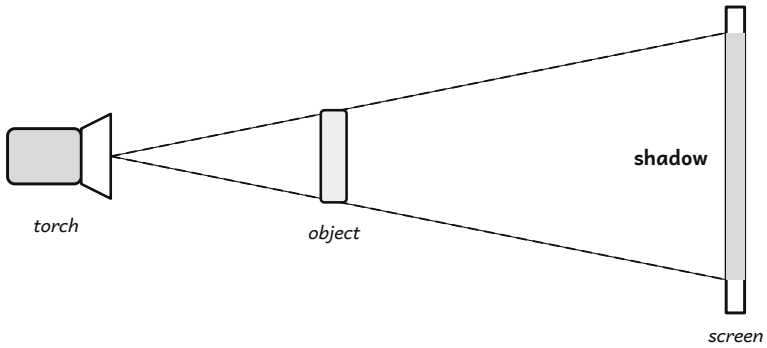
Changing shadows

★★☆
Grade 5

Learning goals

- Explain shadow size and shape.
- Interpret shadow data from a day.

1 **Look at the torch, object and screen.** [4]



a) What would happen to the shadow if you moved the object closer to the torch?

It would get bigger (and a little more blurred).

b) How could you make the shadow a different shape without changing the object?

Turn the object round or tilt it.

c) Why is the shadow the same shape as the object?

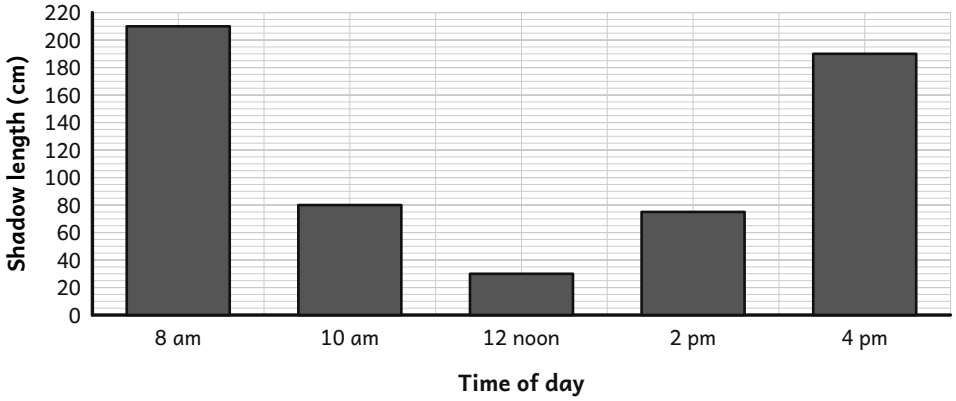
Light travels in straight lines, so it is blocked exactly by the outline of the object.

2 **Why can you not see a clear shadow on a very cloudy day?** [1]

Clouds scatter the sunlight so light comes from all directions; nothing blocks all of it.

3 **Nadia measured the shadow of a 1 metre stick in the school yard.** [6]

Time	8 am	10 am	12 noon	2 pm	4 pm
Shadow length (cm)	210	80	30	75	190



a) Draw a bar chart of the results.

b) Describe the pattern and explain it.

The shadow was long in the morning, shortest at noon and long again in the afternoon, because the Sun is low in the morning and evening and highest at midday.

c) In which direction did the shadow point at 8 am? Why?

West, because the Sun was in the east and shadows point away from the Sun.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · C8 · SHADOWS

How does distance from the torch change shadow size?

★★☆
Grade 5

Learning goals

- Plan a fair test.
- Plot a line graph.
- Use the pattern to predict.

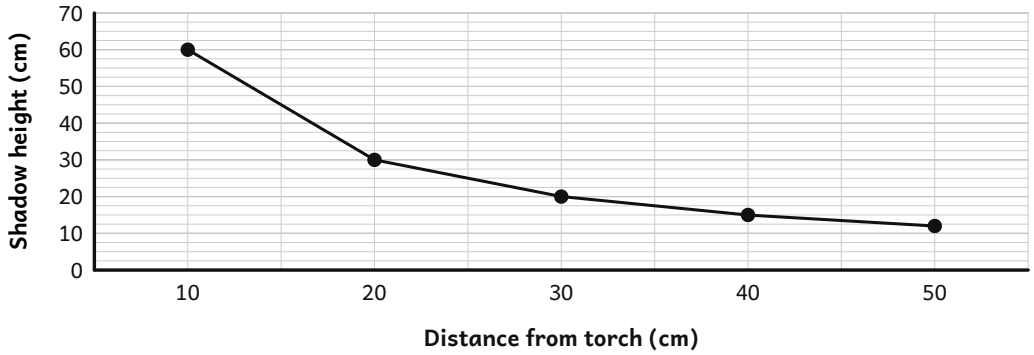
Jamal put a torch on a table and a screen 60 cm away. He held a card puppet 10 cm tall at different distances from the torch and measured the height of the shadow on the screen.

1 **Complete the plan.** [5]

Question	How does the distance between the torch and the puppet change the size of the shadow?
Change	distance of the puppet from the torch
Measure	height of the shadow (cm)
Keep the same	the same puppet and torch, screen distance, room darkness
Prediction	The closer the puppet is to the torch, the bigger the shadow.

2 **Plot the results as a line graph.** [4]

Distance from torch (cm)	10	20	30	40	50
Shadow height (cm)	60	30	20	15	12



3 **Describe the pattern. Did the results support Jamal's prediction?** [2]

The closer the puppet was to the torch, the taller the shadow. Yes, the results support the prediction.

4 **Predict the shadow height when the puppet is 15 cm from the torch.** [1]

about 40 cm (between 60 cm and 30 cm)

5 **Why can the shadow never be smaller than 10 cm in this test?** [1]

Next to the screen, the shadow is the same size as the puppet (10 cm).

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C8 · SHADOWS

A brighter, cooler classroom

★★★
Grade 5

Learning goals

- Use light intensity data.
- Apply knowledge of materials to a design problem.
- Weigh up costs and benefits.

Science | Design | Maths | Environment | Economics

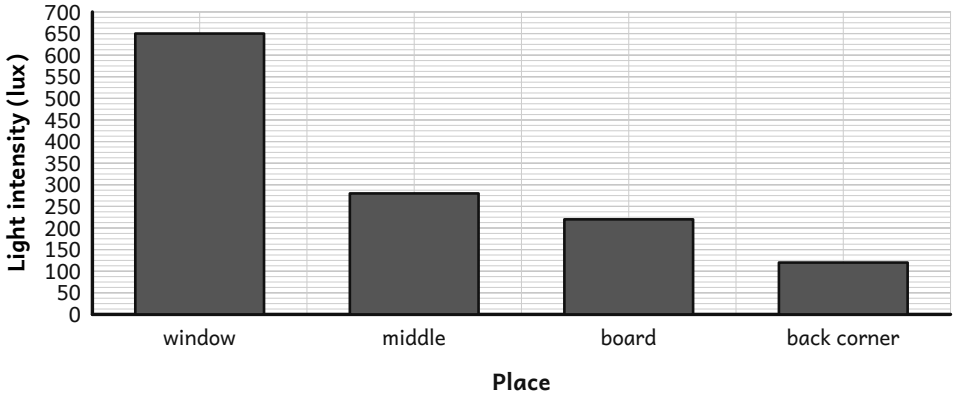
In Bangladesh, classrooms can get very **hot** from April to October. Teachers often close the curtains to keep out the heat and glare, but then the room gets **dark** and the lights must be switched on, which uses electricity. Some learners at the back cannot see the board clearly.

Experts recommend about **300 to 500 lux** for reading and writing in classrooms. Too little light strains the eyes; direct sunlight on desks causes **glare** and heat. Good classroom design uses daylight wisely: light-coloured walls and ceilings **reflect** light around the room, **translucent** blinds let light in but block direct sun, and trees or shades outside block heat from the walls and windows.

Example readings below; lighting levels based on common school lighting guidelines.

1 **Grade 5 measured light intensity around their classroom with a light meter app at 11 am, with the curtains closed and lights off.** [3]

Place	by the window	middle desk	board	back corner
Light (lux)	650	280	220	120



a) Draw the bar chart.

b) Which places are below 300 lux?

middle desk (280), board (220), back corner (120)

2 **For each idea, explain the science.** [3]

Idea	How it helps (use the words reflect, translucent or opaque)
paint the walls white	White surfaces reflect more light, so the room is brighter.
translucent blinds	They let some light in but scatter it, so less glare and heat.
plant trees outside the windows	Opaque leaves block direct sunlight and shade the walls, keeping the room cooler.

3 **What might each person think?** [3]

Person	View
the head teacher	White paint is cheap, but new blinds cost money.
a learner at the back	I need more light to see the board.
the school caretaker	Using daylight saves electricity bills.

4 **Draw and label your design for a brighter, cooler classroom.** [3]

My classroom design

Labels such as white walls and ceiling, translucent blinds, tree shade, windows on two sides, LED lights for cloudy days.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C8 · SHADOWS

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

I	S	O	S	U	N	D	I	A	L	F
T	C	T	U	O	U	Y	F	P	B	C
W	S	K	R	K	I	M	P	T	F	V
N	G	E	G	A	W	V	O	O	N	W
O	U	M	E	S	I	C	R	R	R	O
O	Y	P	A	H	O	G	H	C	D	P
N	I	M	G	A	B	B	H	H	V	A
P	J	E	E	D	L	L	I	T	D	Q
T	A	T	C	O	O	W	K	E	K	U
N	E	E	Y	W	C	R	K	N	R	E
T	P	R	O	M	K	I	T	L	U	X

block lux meter noon opaque shadow straight sundial torch

2 **Which material would make the darkest shadow?** [1]

- ○ clear glass
- ○ tracing paper
- ● **cardboard**
- ○ water

3 **When is your shadow shortest on a sunny day?** [1]

- ○ early morning
- ● **midday**
- ○ late afternoon
- ○ at sunset

4 **Draw the shadow of the stick at 9 am and at 4 pm. Label east and west.** [2]

Stick shadows

9 am: long shadow pointing west (Sun in the east). 4 pm: long shadow pointing east (Sun in the west).

5 **Challenge. How did Ole Rømer show that light takes time to travel? Why was this an important idea?** [2]

Jupiter's moon appeared later when the Earth was further from Jupiter, so the light needed extra time to travel the extra distance. It proved light has a speed and is not instant.